

Quadratic functions - Set 1 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

Answer key

Question	Answer	Topic
1	E	Quadratic functions
2	B	Quadratic functions
3	A	Quadratic functions
4	C	Quadratic functions
5	B	Quadratic functions
6	C	Quadratic functions
7	D	Quadratic functions
8	B	Quadratic functions
9	C	Quadratic functions
10	D	Quadratic functions
11	C	Quadratic functions
12	C	Quadratic functions

Worked solutions

Question 1 QUA-01 · Quadratic functions · spec MM1.3

The equation $x^2 + (k - 2)x + (k + 1) = 0$ has two distinct real roots. Find the complete set of possible values of k .

Answer: E

Discriminant $(k - 2)^2 - 4(k + 1) = k^2 - 8k = k(k - 8) > 0$, so $k < 0$ or $k > 8$.

Question 2 QUA-02 · Quadratic functions · spec MM1.3

By completing the square, find the minimum value of $2x^2 - 12x + 23$.

Answer: B

$2x^2 - 12x + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$, so the minimum is 5 (at $x = 3$).

Question 3 QUA-03 · Quadratic functions · spec MM1.3

The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.

Answer: A

$y = a(x - 2)^2 - 3$; at $x = 0$: $4a - 3 = 5$, so $a = 2$. Then $y = 2(x - 2)^2 - 3 = 2x^2 - 8x + 5$, and $a + b + c = 2 - 8 + 5 = -1$ (which is just $y(1)$).

Question 4 QUA-05 · Quadratic functions · spec MM1.3

For what value of c does the line $y = 2x + c$ touch the parabola $y = x^2 - 4x + 7$?

Answer: C

Equate: $x^2 - 6x + (7 - c) = 0$. Tangency needs discriminant zero: $36 - 4(7 - c) = 0$, so $8 + 4c = 0$ and $c = -2$.

Question 5 QUA-07 · Quadratic functions · spec MM1.3

How many real solutions does the equation $x^4 - 5x^2 + 4 = 0$ have?

Answer: B

Let $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4) = 0$, so $x^2 = 1$ or $x^2 = 4$, giving $x = \pm 1, \pm 2$: four solutions.

Question 6 QUA-08 · Quadratic functions · spec MM1.3

The equation $kx^2 + 4x + k = 0$ has no real roots. Find the complete set of values of k .

Answer: C

Need $16 - 4k^2 < 0$, i.e. $k^2 > 4$, so $k < -2$ or $k > 2$. (If $k = 0$ the equation is linear with one root, so $k = 0$ is excluded anyway.)

Question 7 QUA-10 · Quadratic functions · spec MM1.3

The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0. Which of the following could be the quadratic?

Answer: D

Minimum -4 means $x^2 + px + q = (x - h)^2 - 4$, with roots $h \pm 2$. One root is 0, so $h = \pm 2$: the quadratic is $x^2 - 4x$ or $x^2 + 4x$. Only $x^2 - 4x$ is offered.

Question 8 QUA-11 · Quadratic functions · spec MM1.3

The curve $y = x^2 - 2ax + a^2 + a - 6$ has its vertex on the x -axis. Find the value of a .

Answer: B

Complete the square: $y = (x - a)^2 + a - 6$. The vertex is $(a, a - 6)$, which lies on the x -axis when $a = 6$.

Question 9 QUA-12 · Quadratic functions · spec MM1.3

The roots of $2x^2 + bx + 18 = 0$ are equal. Find the sum of the possible values of b .

Answer: C

$b^2 - 144 = 0$, so $b = \pm 12$; their sum is 0.

Question 10 QUA-13 · Quadratic functions · spec MM1.3

Which of the following quadratics has its vertex in the second quadrant (where $x < 0$ and $y > 0$)?

Answer: D

Complete the square: A vertex $(-2, -3)$; B $(2, 1)$; C $(-1, -2)$; D $(-3, 1)$, which has $x < 0$ and $y > 0$; E $(1, 2)$. So D.

Question 11 QUA-15 · Quadratic functions · spec MM1.3

The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?

Answer: C

$y = (x - 1)(x - 5)$, vertex at $x = 3$: $y = (2)(-2) = -4$.

Question 12 QUA-16 · Quadratic functions · spec MM1.3

Given that $x^2 + y^2 = 10$ and $x + y = 4$, find the value of xy .

Answer: C

$(x + y)^2 = x^2 + y^2 + 2xy$, so $16 = 10 + 2xy$ and $xy = 3$.